Effect of adding public buildings or firms at fixed n and penetration (paired by community)
° = fewer than 80% of communities agree with the sign of the median; 0 = |median| < 0.05

ND1
(320 paired communities)

	Gini of savings	Fairness Index	σ of merit deviation	Max coalition excess	Blocking coalitions	Incentive strength (net)	Alignment (net)	Alignment (gross)
Shapley	-0.1	0	0	0	+0.1	0	-0.1	+0.1°
Variance Least Core	-0.2	0	0	0	0	0	-0.1	0
Equal Split	0	-0.1	-0.1	-0.1	+0.1	0	-0.1°	-0.1
Prop. to Consumption	0	+0.1	0	+0.2	+0.4	-0.1	-0.1	0
Pearson Key	0	0	-0.2	+0.1	-0.1	0	0	-0.3°
Similarity-Utilization	0	-0.1°	-0.3	+0.1°	0	0	0	-0.3
Remuneration Model 1	-0.3	+0.1°	+0.1°	0	0	-0.2	-0.5	+0.1
Cascading Tree	-0.2	+0.1	+0.2	+0.2	+0.1	0	0	+0.5
Weighted Solidarity	0	0	0	0	+0.1	0	-0.1	0
Tri-level EP	0	0	-0.1°	+0.1	0	0	0	-0.2°

ND1_PA1
(320 paired communities)

	Gini of savings	Fairness Index	σ of merit deviation	Max coalition excess	Blocking coalitions	Incentive strength (net)	Alignment (net)	Alignment (gross)
Shapley	-0.1	0	0	0	0	0	-0.1	+0.1°
Variance Least Core	-0.2	0	0	0	0	0	-0.1°	-0.2°
Equal Split	0	-0.1	0	-0.2	0	+0.1	-0.1°	-0.1°
Prop. to Consumption	0	+0.1	+0.1	+0.2	+0.2	-0.1	-0.1	-0.2°
Pearson Key	+0.1	+0.1	0	+0.1	+0.1°	0	0	-0.4
Similarity-Utilization	0	-0.1°	-0.1°	+0.1°	-0.1	0	0	-0.5
Remuneration Model 1	-0.5	-0.3	-0.5	-0.2	-0.1°	-0.3	-0.6	+0.1
Cascading Tree	-0.3	0	0	+0.2°	0	0	0	+0.6
Weighted Solidarity	0	0	0	0	0	+0.1	-0.1	+0.1
Tri-level EP	+0.1	+0.1	+0.1°	+0.2	+0.1	-0.1	0	-0.4

ND1_PA2
(320 paired communities)

	Gini of savings	Fairness Index	σ of merit deviation	Max coalition excess	Blocking coalitions	Incentive strength (net)	Alignment (net)	Alignment (gross)
Shapley	-0.2	+0.1	+0.1°	0	+0.1°	-0.2	-0.4°	+0.1°
Variance Least Core	-0.1	0	0	0	0	-0.3	-0.5°	+0.1
Equal Split	0	-0.1	-0.1	-0.1	+0.1	0	-0.2	-0.1°
Prop. to Consumption	-0.2	0	-0.1°	+0.1°	-0.2°	-0.1°	-0.6°	-0.2°
Pearson Key	0	+0.1°	0	+0.2	+0.3	-0.2	-0.5°	-0.2
Similarity-Utilization	+0.2	-0.2°	-0.1°	-0.1°	+0.2	0	0	-0.4
Remuneration Model 1	-0.4	+0.3	+0.3	+0.2	+0.5°	-0.3°	-0.6°	+0.1
Cascading Tree	-0.3	+0.5	+0.5	+0.5	+0.4	+0.1	0	+0.6
Weighted Solidarity	-0.1	+0.6	+0.7	+0.6	+0.5	+0.2	+0.1°	+0.5
Tri-level EP	0	+0.1°	0	+0.2	0	-0.2	-0.5°	-0.2

ND1_PA3
(320 paired communities)

	Gini of savings	Fairness Index	σ of merit deviation	Max coalition excess	Blocking coalitions	Incentive strength (net)	Alignment (net)	Alignment (gross)
Shapley	-0.2	+0.1	+0.1	0	+0.1°	-0.1°	-0.4	+0.2
Variance Least Core	-0.2	0	-0.1	0	0	-0.3	-0.6	+0.1
Equal Split	0	-0.1	-0.1	-0.2	0	+0.2	-0.1	-0.2
Prop. to Consumption	-0.2	+0.2	+0.2	+0.2	+0.2°	-0.3	-0.6	0
Pearson Key	0	+0.2°	0	+0.2	+0.4	-0.2°	-0.5	0
Similarity-Utilization	+0.3	-0.2°	-0.2	-0.1°	-0.1	0	0	-0.4
Remuneration Model 1	-0.5	+0.4	+0.4	+0.3	+0.6	-0.1°	-0.6	0
Cascading Tree	-0.4	+0.4	+0.2	+0.4	0	0	0	0
Weighted Solidarity	-0.2	+0.7	+0.7	+0.6	+0.4	+0.4	+0.2°	+0.7
Tri-level EP	0	+0.2°	+0.1°	+0.2	+0.4	-0.2	-0.5	0

ND2
(320 paired communities)

	Gini of savings	Fairness Index	σ of merit deviation	Max coalition excess	Blocking coalitions	Incentive strength (net)	Alignment (net)	Alignment (gross)
Shapley	-0.1°	0	0	0	0	0	-0.1	+0.1°
Variance Least Core	-0.1	0	0	0	0	0	-0.1°	-0.2°
Equal Split	0	0	+0.2	-0.1	+0.1	+0.1	-0.1°	-0.1°
Prop. to Consumption	0	+0.1°	+0.1°	+0.1°	+0.2°	-0.2	-0.3	+0.2
Pearson Key	+0.2	+0.2	+0.1°	+0.2	+0.4	-0.1	-0.1	-0.4°
Similarity-Utilization	+0.1	-0.1°	0	+0.1°	+0.1°	0	0	-0.4
Remuneration Model 1	-0.6	-0.4	-0.5	-0.4	-0.1	-0.3	-0.6	0
Cascading Tree	-0.5	0	0	0	+0.1°	0	0	+0.5
Weighted Solidarity	+0.1°	0	+0.2	0	+0.1	+0.1	-0.1	+0.2
Tri-level EP	+0.2	+0.2	+0.2	+0.2	+0.5	-0.1	-0.1	-0.3°

ND2_PA1
(320 paired communities)

	Gini of savings	Fairness Index	σ of merit deviation	Max coalition excess	Blocking coalitions	Incentive strength (net)	Alignment (net)	Alignment (gross)
Shapley	0	0	0	-0.1	0	0	-0.1	+0.1°
Variance Least Core	0	0	0	0	0	0	-0.1°	-0.3°
Equal Split	0	0	+0.2	-0.1°	0	+0.2	-0.1°	-0.1°
Prop. to Consumption	0	0	-0.1°	0	0	-0.2	-0.3	0
Pearson Key	+0.2	+0.2	+0.1	+0.2	+0.3	-0.1	-0.1	-0.3
Similarity-Utilization	+0.1	-0.1°	0	0	0	0	0	-0.5
Remuneration Model 1	-0.6	-0.5	-0.5	-0.5	-0.1	-0.4	-0.6	+0.1
Cascading Tree	-0.5	0	0	0	0	0	0	+0.6
Weighted Solidarity	0	0	+0.2	0	+0.1	+0.1	-0.1	+0.4
Tri-level EP	+0.2	+0.3	+0.2	+0.2	+0.3	-0.1	-0.1	-0.3

ND2_PA2
(320 paired communities)

	Gini of savings	Fairness Index	σ of merit deviation	Max coalition excess	Blocking coalitions	Incentive strength (net)	Alignment (net)	Alignment (gross)
Shapley	-0.2	0	0	0	-0.2	-0.1°	-0.4°	0
Variance Least Core	-0.2	0	0	0	0	-0.1°	-0.4°	-0.1°
Equal Split	0	-0.1	-0.1	-0.2	0	0	-0.2	-0.1°
Prop. to Consumption	-0.1	+0.2°	+0.1°	+0.2	0	-0.1°	-0.5°	-0.1°
Pearson Key	0	+0.2°	+0.1°	+0.2	+0.2	-0.1	-0.5°	-0.2°
Similarity-Utilization	+0.2	-0.2°	-0.1°	-0.1°	-0.1	0	0	-0.5
Remuneration Model 1	-0.5	-0.1	-0.1	0	-0.1°	-0.4	-0.7	+0.1
Cascading Tree	-0.4	+0.3	+0.2	+0.4	0	+0.1	0	+0.6
Weighted Solidarity	-0.2	+0.6	+0.7	+0.6	+0.3	+0.3	+0.1°	+0.5
Tri-level EP	0	+0.2°	+0.1°	+0.2	-0.1°	-0.1	-0.5°	-0.2°

ND2_PA3
(300 paired communities)

	Gini of savings	Fairness Index	σ of merit deviation	Max coalition excess	Blocking coalitions	Incentive strength (net)	Alignment (net)	Alignment (gross)
Shapley	-0.3	+0.1	+0.1	0	0	-0.2°	-0.6	+0.2
Variance Least Core	-0.2	0	0	0	0	-0.4	-0.6	0
Equal Split	0	-0.1	-0.1	-0.2	+0.2°	+0.1	-0.1	-0.1°
Prop. to Consumption	-0.2	+0.2	+0.2	+0.2	+0.3	-0.3°	-0.6	0
Pearson Key	0	+0.2	+0.1°	+0.2	+0.5	-0.3	-0.5	0
Similarity-Utilization	+0.3	-0.4°	-0.4°	-0.2	0	0	0	-0.4
Remuneration Model 1	-0.5	+0.4	+0.3	+0.2	+0.6	-0.2°	-0.6	+0.1
Cascading Tree	-0.5	+0.4	+0.3	+0.5	0	0	0	+0.3
Weighted Solidarity	-0.2	+0.7	+0.7	+0.6	+0.5	+0.3	+0.2°	+0.8
Tri-level EP	0	+0.2	+0.1°	+0.2	+0.6	-0.3	-0.5	0

ND3
(320 paired communities)

	Gini of savings	Fairness Index	σ of merit deviation	Max coalition excess	Blocking coalitions	Incentive strength (net)	Alignment (net)	Alignment (gross)
Shapley	-0.1	+0.1	+0.1	0	0	-0.1°	-0.3	+0.2
Variance Least Core	-0.1	0	0	0	0	-0.2°	-0.3	+0.1°
Equal Split	0	-0.1	0	-0.1	+0.2°	+0.1	-0.1°	-0.1°
Prop. to Consumption	-0.1°	+0.1°	0	0	-0.2°	-0.4°	-0.5	+0.2
Pearson Key	+0.1	+0.1°	-0.1°	+0.1	0	-0.2	-0.3	0
Similarity-Utilization	+0.2	-0.3	-0.3	-0.2°	0	0	0	-0.4
Remuneration Model 1	-0.5	0	-0.3	-0.2°	0	-0.4	-0.6	0
Cascading Tree	-0.4	+0.4	+0.3	+0.3°	+0.1°	0	0	+0.6
Weighted Solidarity	0	0	+0.2	0	+0.2	+0.1	-0.1°	+0.4
Tri-level EP	+0.1	+0.1°	-0.1°	+0.1	0	-0.3	-0.3	0

ND3_PA1
(320 paired communities)

	Gini of savings	Fairness Index	σ of merit deviation	Max coalition excess	Blocking coalitions	Incentive strength (net)	Alignment (net)	Alignment (gross)
Shapley	-0.1	0	+0.1	0	0	-0.1°	-0.3	+0.2
Variance Least Core	0	0	0	0	0	-0.2°	-0.2	0
Equal Split	0	0	0	-0.1	+0.1°	+0.1	0	0
Prop. to Consumption	-0.1°	+0.1°	+0.1°	-0.1°	-0.1	-0.4°	-0.5	+0.1°
Pearson Key	+0.1	+0.1°	+0.1°	+0.1	+0.1°	-0.2	-0.3	-0.1°
Similarity-Utilization	+0.2	-0.3	-0.3°	-0.3	-0.1	0	0	-0.4
Remuneration Model 1	-0.5	-0.2	-0.4	-0.2°	-0.1	-0.4	-0.6	+0.1
Cascading Tree	-0.4	+0.3°	+0.2	+0.2°	0	+0.1	0	+0.6
Weighted Solidarity	0	0	+0.2	0	+0.1°	+0.1	-0.1°	+0.5
Tri-level EP	0	+0.1°	0	+0.1	0	-0.3	-0.3	-0.1°

ND3_PA2
(300 paired communities)

	Gini of savings	Fairness Index	σ of merit deviation	Max coalition excess	Blocking coalitions	Incentive strength (net)	Alignment (net)	Alignment (gross)
Shapley	-0.2	0	0	0	-0.2°	-0.2°	-0.6°	0
Variance Least Core	-0.1	0	0	0	0	-0.2°	-0.6°	-0.1°
Equal Split	0	-0.1	-0.1	-0.1	0	0	-0.1	-0.1°
Prop. to Consumption	-0.1	0	0	+0.1	+0.1°	-0.2°	-0.6°	0
Pearson Key	0	+0.1°	+0.1°	+0.2	+0.2	-0.3	-0.5°	-0.2°
Similarity-Utilization	+0.2	-0.4°	-0.4°	-0.2°	-0.1	0	0	-0.5
Remuneration Model 1	-0.5	0	-0.1	0	-0.1°	-0.4	-0.7	+0.1
Cascading Tree	-0.4	+0.3	+0.2	+0.4	0	+0.1	0	+0.6
Weighted Solidarity	-0.2	+0.6	+0.7	+0.6	+0.4	+0.3	+0.2°	+0.6
Tri-level EP	0	+0.1°	0	+0.1	-0.1	-0.3	-0.5°	-0.2°

ND3_PA3
(260 paired communities)

	Gini of savings	Fairness Index	σ of merit deviation	Max coalition excess	Blocking coalitions	Incentive strength (net)	Alignment (net)	Alignment (gross)
Shapley	-0.3	0	+0.1	0	+0.2°	-0.3°	-0.6	+0.2
Variance Least Core	-0.1	0	0	0	0	-0.4	-0.7	+0.1°
Equal Split	0	-0.1	-0.1	-0.1	+0.3	+0.1	-0.1	-0.1°
Prop. to Consumption	-0.2	+0.2	+0.2	+0.2	+0.4	-0.3°	-0.6	+0.1°
Pearson Key	0	+0.1°	0	+0.1	+0.4	-0.3	-0.5	+0.1°
Similarity-Utilization	+0.3	-0.5°	-0.4	-0.3	0	0	0	-0.3
Remuneration Model 1	-0.5	+0.4	+0.3	+0.3	+0.7	-0.2°	-0.6	+0.1
Cascading Tree	-0.5	+0.5	+0.3	+0.6	0	0	0	+0.3
Weighted Solidarity	-0.3	+0.7	+0.7	+0.7	+0.5	+0.3	+0.2	+0.8
Tri-level EP	0	+0.1°	0	+0.1	+0.6	-0.3	-0.5	+0.1°

PA1: + town hall
(560 paired communities)

	Gini of savings	Fairness Index	σ of merit deviation	Max coalition excess	Blocking coalitions	Incentive strength (net)	Alignment (net)	Alignment (gross)
Shapley	-0.1	+0.1	+0.1°	0	0	0	-0.1	+0.2°
Variance Least Core	-0.2	0	0	0	0	0	-0.1°	0
Equal Split	0	-0.1	0	-0.2	+0.2	0	-0.1°	-0.1
Prop. to Consumption	0	0	-0.1	+0.2	0	0	0	-0.2°
Pearson Key	+0.1	0	-0.2	+0.1	-0.1	0	0	-0.4
Similarity-Utilization	0	0	-0.2	+0.1°	+0.1°	0	0	-0.4
Remuneration Model 1	-0.5	-0.1°	-0.2°	-0.1°	-0.1°	-0.3	-0.6	+0.2
Cascading Tree	-0.3	+0.1	+0.2	+0.2	+0.2	0	0	+0.5
Weighted Solidarity	0	0	0	0	+0.2	0	-0.1	0
Tri-level EP	+0.1	0	-0.2	+0.1	-0.1°	0	0	-0.4

PA2: + town hall, school
(540 paired communities)

	Gini of savings	Fairness Index	σ of merit deviation	Max coalition excess	Blocking coalitions	Incentive strength (net)	Alignment (net)	Alignment (gross)
Shapley	-0.2	+0.1	0	0	+0.2	-0.3	-0.4	+0.2
Variance Least Core	-0.1°	0	-0.1	0	0	-0.4	-0.6	+0.2
Equal Split	0	-0.1	-0.1	-0.1	+0.2	0	-0.2	-0.2
Prop. to Consumption	-0.1	-0.1°	-0.4	+0.1°	-0.4°	-0.1°	-0.6°	-0.1°
Pearson Key	0	-0.1°	-0.3°	+0.2	+0.1°	-0.2	-0.5°	-0.2°
Similarity-Utilization	+0.3	-0.3°	-0.3°	0	+0.3°	0	0	-0.4
Remuneration Model 1	-0.4	+0.4	+0.4	+0.3	+0.4	-0.3°	-0.6	+0.1
Cascading Tree	-0.3	+0.5	+0.6	+0.5	+0.4°	+0.1	0	+0.5
Weighted Solidarity	-0.2	+0.5	+0.5	+0.6	+0.3	+0.3	+0.1°	+0.4
Tri-level EP	0	-0.1°	-0.3°	+0.2	+0.1°	-0.2	-0.5°	-0.2°

PA3: + town hall, school, sports hall
(500 paired communities)

	Gini of savings	Fairness Index	σ of merit deviation	Max coalition excess	Blocking coalitions	Incentive strength (net)	Alignment (net)	Alignment (gross)
Shapley	-0.2	+0.1	+0.1	0	+0.1°	-0.1°	-0.5	+0.2
Variance Least Core	-0.1	-0.1	-0.1	0	0	-0.4	-0.6	+0.1
Equal Split	0	-0.2	-0.1	-0.2	0	+0.1	-0.1	-0.2
Prop. to Consumption	-0.2	+0.1	+0.1°	+0.2	0	-0.2	-0.6	+0.1°
Pearson Key	0	0	-0.2°	+0.2	+0.1	-0.1°	-0.4	+0.1°
Similarity-Utilization	+0.3	-0.4	-0.4	-0.1°	-0.2	0	0	-0.1°
Remuneration Model 1	-0.5	+0.4	+0.4	+0.3	+0.6	0	-0.5	0
Cascading Tree	-0.5	+0.4	+0.2°	+0.5	0	0	0	-0.1°
Weighted Solidarity	-0.2	+0.7	+0.7	+0.6	+0.4	+0.4	+0.2	+0.7
Tri-level EP	0	0	-0.1°	+0.2	+0.1°	-0.1	-0.5	+0.1°

median Δ score vs households only
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00